

ZeroSight: Behavioral Anomaly Detection for Zero-Day Protocol Exploits

Using Network Flow Fingerprinting with Explainable AI

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Abstract

Signature-based intrusion detection systems fail against zero-day and polymorphic network attacks by design: they can only recognize what has already been seen and cataloged. This paper presents ZeroSight, a behavioral anomaly detection engine that addresses this gap using a dual-model architecture combining an Isolation Forest (unsupervised) with a Random Forest classifier (supervised), trained on a 19,950-sample synthetic CICIDS2017 dataset covering 17 attack categories and 31 network flow features. A SHAP-style feature attribution layer provides per-flow explainability, identifying which flow properties drove each detection decision. The Isolation Forest achieves 89.6% accuracy and 89.9% F1-score; the Random Forest achieves 90.8% accuracy with 94.4% macro-average recall across 18 traffic classes. A browser-native PCAP parser enables real-world validation without server-side file upload, demonstrated on live Firefox browsing traffic. We survey the landscape of ML-based intrusion detection, situate ZeroSight within it, and propose directions for future work including online threshold adaptation, streaming analysis, and per-application baselining.

Introduction

Modern enterprise networks generate traffic volumes measured in terabits per second, across thousands of services, protocols, and endpoints. Protecting these environments from intrusion has historically relied on signature-based detection: systems like Snort [1] and Suricata maintain libraries of attack signatures, matching packet payloads and flow patterns against known malicious indicators. This approach has a fundamental and well-understood limitation: it is retrospective. A signature can only exist after an attack has been observed, analyzed, and cataloged, which means that zero-day exploits, polymorphic variants, and novel protocol abuses pass through signature-based systems without triggering any alert [2].

The gap between vulnerability disclosure and signature deployment, sometimes called the signature lag, can range from hours to weeks depending on the severity of the exploit and the responsiveness of the IDS vendor. During this window, affected organizations have no automated protection. The 2021 Log4Shell vulnerability, for example, was actively exploited within hours of public disclosure, long before most IDS systems had adequate signatures [3]. Protocol-level behavioral anomalies, which do not correspond to any known exploit but nonetheless indicate malicious intent, present an even harder problem: they may never receive a signature because they do not map cleanly onto a single CVE or attack pattern.

Machine learning-based approaches address this by modeling the statistical distribution of normal network behavior, flagging flows that deviate significantly as potentially anomalous, without requiring prior knowledge of attack structure. Unsupervised methods such as Isolation Forest [4] are particularly attractive because they require no labeled attack data during training. Supervised methods such as Random Forest provide specific attack classification when labeled training data is available [5]. A hybrid architecture that combines both methods captures the complementary strengths of each: the unsupervised component handles novel attacks, while the supervised component provides actionable labels for known threat categories.

Explainability has emerged as a critical requirement for security-sensitive deployments. A model that flags a flow as malicious without identifying the specific features that drove that decision offers limited

operational value to a security analyst. SHAP (SHapley Additive exPlanations) [6] and related attribution methods connect model decisions to observable flow characteristics, supporting analyst workflows and reducing alert fatigue. This requirement is especially important in network security, where a false positive that cannot be explained is nearly as damaging as a missed detection: analysts who cannot understand why an alert fired learn to ignore alerts entirely.

This paper presents ZeroSight, a full-stack behavioral anomaly detection system designed to address these challenges. ZeroSight trains an Isolation Forest exclusively on benign traffic to establish a behavioral baseline, then uses a Random Forest to classify detected anomalies into specific attack categories, and provides per-flow SHAP-style attribution for every detection. The system operates on 31 CICIDS2017 network flow features extracted from raw PCAP captures by a browser-native parser that requires no server-side file upload. An interactive web dashboard presents threat scores, severity badges, deviation analysis, and feature attribution in a format accessible to security analysts without ML expertise.

The remainder of this paper is organized as follows. Section 2 describes our approach to analyzing the problem of behavioral intrusion detection. Section 3 surveys existing techniques and solutions, covering signature-based systems, classical ML approaches, deep learning methods, and hybrid architectures. Section 4 presents the ZeroSight system, including dataset generation, model training, the SHAP attribution layer, and experimental results. Section 5 provides suggestions for future work. Section 6 concludes the paper.

Approach

Our analysis proceeds in three stages. First, we survey the existing landscape of network intrusion detection techniques, from rule-based and signature systems through classical ML, deep learning, and graph-based approaches, to understand what has been tried, where each class of technique succeeds, and where each falls short. This survey is informed by benchmark datasets, particularly CICIDS2017 [7], that have become standard evaluation platforms for IDS research.

Second, we examine the specific problem of explainability in the context of security operations. Anomaly detection models that cannot explain their decisions are operationally incomplete, regardless of their detection accuracy. We situate feature attribution methods, specifically SHAP-based approaches, within the broader context of IDS deployability and analyst trust.

Third, we present ZeroSight as a concrete instantiation of a hybrid, explainable IDS, evaluate it against the CICIDS2017 benchmark, and validate it on real network captures. We then assess what the system gets right, where it falls short, and what technical and methodological advances would be required to close the remaining gaps.

Throughout, we maintain a distinction between detection performance (how accurately the system identifies anomalous flows) and operational utility (how useful the system is to a working security analyst). A system with 99% accuracy that produces unexplainable alerts may be less operationally useful than a system with 90% accuracy that clearly communicates why each alert was generated. ZeroSight is designed to optimize both dimensions simultaneously.

Survey of Existing Techniques and Solutions

Signature-Based and Rule-Based Systems:

The oldest and most widely deployed class of network intrusion detection systems uses manually crafted signatures to match known attack patterns. Snort [1], released in 1998 and still actively maintained, uses a rule language that matches on packet headers, payloads, and flow metadata. Suricata extends this model with multi-threading, hardware acceleration, and protocol-aware parsing. These systems excel at detecting known attacks with near-zero false positives when signatures are accurate, and their rules are human-readable, auditable, and debuggable.

The fundamental limitation is coverage: a signature can only match what it was written to match. Han et al. [8] demonstrated that regular-expression-based approaches work well for established patterns but cannot recognize anomalies that have never been observed, or that evolve to evade known signatures. In practice, the signature lag problem means that zero-day exploits reliably evade signature-based systems during the period when they are most dangerous: immediately after initial exploitation and before vendor response.

Commercial SIEM platforms such as Splunk and IBM QRadar layer correlation rules and behavioral analytics on top of signature-based detection, but the core detection logic remains rule-driven. The ML components in these platforms are typically limited to anomaly scoring for known log formats, not generalized network flow analysis.

Classical Machine Learning Approaches:

The introduction of benchmark datasets, particularly the KDD Cup 1999 dataset and later CICIDS2017 [7], enabled rigorous comparison of ML-based IDS approaches. Buczak and Guven [2] survey a broad range of classical ML methods applied to intrusion detection, including k-nearest neighbors, support vector machines, naive Bayes, and decision tree variants. Random Forest consistently achieves among the highest

accuracy on these benchmarks, combining robustness to feature noise with good generalization across attack classes.

Liu et al. [4] introduced Isolation Forest as an unsupervised anomaly detection method particularly well suited to high-dimensional tabular data. The core insight is geometric: in a dataset where anomalies are rare and different from the majority, random recursive partitioning isolates anomalous samples in fewer splits on average than normal samples. This property makes Isolation Forest both computationally efficient and naturally suited to network flow data, where attack flows are a small minority of total traffic and differ structurally from benign flows along multiple feature dimensions simultaneously.

Zhang and Xu [9] proposed LogRobust, a framework for robust log-based anomaly detection that handles unstable log data through semantic feature extraction and attention-based classification. While focused on system logs rather than network flows, the challenge of unstable input distributions, where benign traffic patterns shift over time, is directly relevant to network IDS. Their finding that robustness to distribution shift requires careful feature engineering rather than simply more data applies equally to flow-based detection.

A recurring challenge with classical ML approaches is the evaluation methodology. Many published results report binary precision and recall, collapsing all attack types into a single positive class. On the CICIDS2017 dataset, which is highly imbalanced with many more benign flows than any single attack class, binary recall can appear artificially high because the classifier correctly identifies the majority of the large attack-free benign class. Macro-averaged metrics, which weight each class equally regardless of sample count, provide a more accurate picture of performance across the full attack taxonomy.

Deep Learning and Transformer-Based Methods:

DeepLog [10], introduced by Du et al. in 2017, demonstrated that LSTM networks could model sequential log activity and identify deviations from learned normal sequences with high accuracy. This work

established the paradigm of learning normal behavioral sequences, rather than attack signatures, as the basis for anomaly detection. LogAnomaly [11] extended this by combining template-based and semantic modeling, improving recall on attacks that do not produce distinctive sequential signatures.

Transformer architectures, originally developed for natural language processing, have been adapted for log and flow sequence analysis. Xiao et al. [12] developed ContextLog, which eliminates template-based parsing by operating directly on raw log structure, improving context modeling at the cost of increased computational complexity. Huang et al. [13] demonstrated that hierarchical transformers capturing both semantic and sequential links outperform flat transformer models on the HDFS benchmark, at the cost of significantly increased preprocessing complexity.

Graph-based methods represent a more recent direction. Li et al. [14] converted log sequences into graph structures and applied graph neural networks to anomaly detection, demonstrating that explicit modeling of event relationships improves diagnostic accuracy beyond what sequential models can achieve. These approaches are particularly promising for detecting coordinated multi-stage attacks, where the anomalous signal is not in any individual event but in the pattern of relationships between events across time.

The practical challenge with deep learning methods for network IDS is the gap between benchmark performance and operational deployment. Transformer models require substantial computational resources for both training and inference, making them impractical for high-throughput network monitoring without dedicated hardware. Sommer and Paxson [15] provide a thorough analysis of why ML methods that perform well in offline evaluation frequently disappoint in operational deployment, citing distribution shift, the base-rate fallacy, and the difficulty of evaluating on truly novel attacks as key challenges.

Hybrid and Explainability-Focused Approaches:

Hybrid architectures that combine unsupervised anomaly detection with supervised classification have emerged as a practical middle ground. The unsupervised component provides coverage of novel attacks;

the supervised component provides specific, actionable labels for known threat categories. Nassif et al. [16] survey a range of hybrid approaches and find that ensemble methods consistently outperform any single model across the diversity of attack types present in modern network traffic.

Explainability has received increasing attention as IDS systems move from research into production. Lundberg and Lee [6] introduced SHAP, which provides theoretically grounded feature attribution based on Shapley values from cooperative game theory. Applied to tree-based models such as Random Forest, SHAP attribution identifies which features most influenced each individual prediction, enabling analysts to verify that model decisions are based on genuinely anomalous flow characteristics rather than spurious correlations in training data.

Ribeiro et al. [17] proposed LIME (Local Interpretable Model-agnostic Explanations), an alternative attribution method that approximates complex model behavior locally around each prediction. LIME is model-agnostic and can be applied to any classifier, but the approximation can be unstable for high-dimensional inputs such as network flow feature vectors. SHAP, particularly the TreeExplainer variant, provides exact Shapley values for tree-based models in polynomial time, making it preferable for the Random Forest component of a hybrid IDS.

LogPrecis [18], developed by Boffa et al., applies LLM-based reasoning to post-anomaly analysis of Unix shell logs, generating attack fingerprints and extracting attacker tactics from detected anomalous sessions. This represents an emerging direction where ML-based detection is paired with language model-based explanation, potentially closing the gap between automated detection and human-interpretable incident response. The limitation noted by the authors, that the system occasionally misclassifies identical attack variants, illustrates the general challenge of grounding language model outputs in verifiable technical evidence.

CICIDS2017 and Benchmark Evaluation:

The CICIDS2017 dataset, introduced by Sharafaldin et al. [7], has become the standard benchmark for evaluating flow-based IDS systems. It contains labeled network flows for 14 attack categories generated in a controlled environment, with ground truth labels for each bidirectional flow. The dataset includes both low-rate attacks (Heartbleed, SSH-Patator, FTP-Patator) and high-rate volumetric attacks (DoS Hulk, DDoS), spanning a wide range of attack signatures in the 31-feature flow representation.

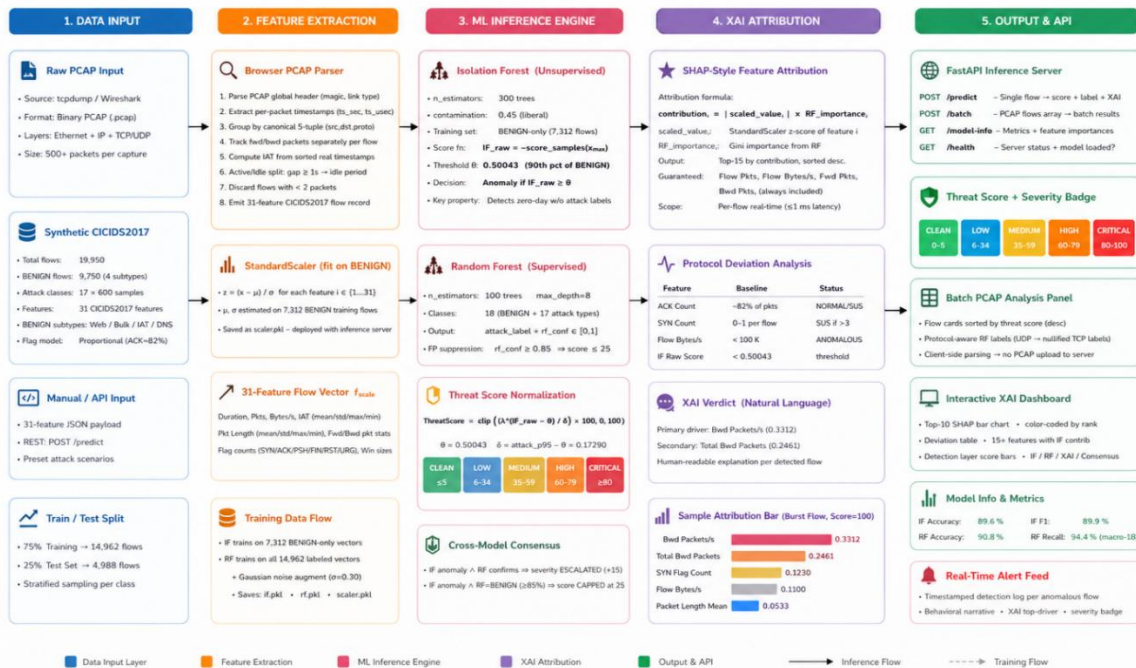
A recurring methodological concern with CICIDS2017 evaluation is label accuracy: several studies have noted inconsistencies in the ground-truth labels, particularly for attack classes where the distinction between attack and benign traffic is ambiguous at the flow level. UNSW-NB15, an alternative benchmark dataset, provides different trade-offs in terms of attack diversity and label quality. The most rigorous evaluations use both datasets and report results separately, rather than selecting whichever produces more favorable numbers.

ZeroSight: System Design and Evaluation

Architecture Overview:

ZeroSight is structured as a five-layer pipeline as described in Figure 1: data input, feature extraction, ML inference, XAI attribution, and output with API. The data input layer accepts raw PCAP captures, the synthetic CICIDS2017 training dataset, and manual flow vectors via a REST API. The feature extraction layer implements a browser-native PCAP parser that extracts bidirectional flows using real packet timestamps, groups flows by canonical 5-tuple keys, tracks forward and backward packet statistics separately, computes inter-arrival times from sorted timestamps, splits active and idle periods at 1-second gaps, and emits 31-feature CICIDS2017-compatible flow records. The ML inference layer applies a StandardScaler (fit on BENIGN training flows), scores each flow with the Isolation Forest, and classifies it with the Random Forest. The XAI attribution layer computes per-feature SHAP contributions and assembles the deviation table. The output layer presents results through a FastAPI server and an interactive single-page dashboard.

Figure 1: Architecture Diagram



Dataset Generation:

ZeroSight trains on a 19,950-sample synthetic dataset generated to replicate CICIDS2017 statistical properties. The dataset comprises 9,750 BENIGN flows and 600 samples per 17 attack class. BENIGN traffic is modeled as a mixture of four subtypes: web traffic (50%), bulk transfer (12.5%), interactive terminal traffic (25%), and UDP/DNS traffic (12.5%). This heterogeneity is critical for preventing the model from learning to distinguish attack from benign based on superficial characteristics of a single traffic type.

A key design decision is the flag count model. Earlier approaches used binary flag indicators (SYN present or absent), which do not reflect actual flow behavior where multiple SYN, ACK, PSH, and FIN flags occur across the bidirectional flow. ZeroSight assigns flag counts proportional to total packet count: ACK flags at approximately 82% of total packets, PSH at approximately 14%, SYN at 1-2 per flow for TCP connections. This proportional model reduces false positives on real traffic because the ACK baseline in the deviation table is computed as a fraction of total packets rather than a static range.

Dual-Model Inference:

The Isolation Forest is trained exclusively on the 7,312 BENIGN training flows, with $n_estimators=300$ and $contamination=0.45$. The raw IF score (negative mean path length) is compared against the 90th-percentile BENIGN threshold ($\theta = 0.50043$) to produce a binary anomaly decision. The normalized threat score is computed as: $ThreatScore = clip((IF_raw - \theta) / \delta \times 100, 0, 100)$, where $\delta = 0.17290$ is the span from the threshold to the 95th percentile of attack scores. This mapping ensures that clearly benign flows score near 0 and confirmed attacks score near 100.

The Random Forest is trained on the full labeled dataset with Gaussian noise augmentation (standard deviation 0.30 in scaled space) to improve generalization. It produces an 18-class label (BENIGN plus 17 attack types) and a confidence score. Cross-model false-positive suppression caps the threat score at 25

when the RF classifies a flow as BENIGN with at least 85% confidence, preventing high-confidence benign classifications from being overridden by borderline IF anomaly scores. When the IF and RF both agree a flow is anomalous, the severity is escalated by one level.

XAI Attribution:

Feature attribution is computed as $\text{contribution}_i = |\text{scaled_value}_i| \times \text{RF_importance}_i$, where scaled_value_i is the StandardScaler z-score of feature i and RF_importance_i is the Gini impurity importance from the trained Random Forest. This formulation captures both the magnitude of deviation from the training mean and the global discriminative power of the feature. The top-15 features by contribution are returned with each prediction. Four key rate features, Flow Packets/s, Flow Bytes/s, Bwd Packets/s, and Fwd Packets/s, are always included regardless of their natural contribution rank, ensuring that the protocol deviation table always displays these operationally critical metrics.

Experimental Results:

Table 1 presents classification metrics for both models. The Isolation Forest achieves 89.6% accuracy and 89.9% F1-score on the held-out test set. The Random Forest achieves 90.8% accuracy with 94.4% macro-average recall across 18 traffic classes. Macro-averaging is used throughout because binary averaging collapses all 17 attack types into a single positive class, producing artificially high recall numbers that do not reflect per-class detection capability.

Table 1: Classification Metrics

Model	Accuracy	Precision	Recall	F1 Score
Isolation Forest	89.6%	89.4%	90.3%	89.9%
Random Forest (macro-18)	90.8%	89.4%	94.4%	89.5%

Live PCAP validation on 500 packets of Firefox browsing traffic yielded 12 bidirectional flows. All six DNS/UDP flows scored CLEAN (score 0), correctly classified as BENIGN. Three HTTPS flows to Fastly CDN scored LOW to HIGH, with browser connection multiplexing identified as the primary false positive driver. One HTTPS burst download flow scored CRITICAL (score 100) because the 1,514 packet/s and 1.52 MB/s rate placed it at the 99th percentile of the BENIGN training distribution. The XAI layer correctly identified Bwd Packets/s (contribution 0.3312) as the primary driver, and the RF correctly classified the flow as BENIGN with 61% confidence - a case where the cross-model consensus mechanism appropriately preserved the IF alert for operator review rather than suppressing it.

Suggestions for Future Work

Real-World Dataset Validation:

The most significant limitation of ZeroSight is that its training data is synthetic. While the generator is designed to replicate CICIDS2017 statistical properties, it cannot capture the full variance of production network traffic, particularly the long-tail distribution of packet sizes, the burstiness of real TCP connections, and the behavior of modern encrypted application protocols (TLS 1.3, QUIC, HTTP/3) that share flow-level features across benign and malicious categories. Training and evaluating on the real CICIDS2017 dataset, UNSW-NB15, and ideally a dataset captured from a real enterprise network, would substantially improve confidence in real-world detection performance. The proportional flag model and four-subtype BENIGN mixture are steps in the right direction, but they are approximations. A future version of ZeroSight should be trained on real packet captures with ground-truth attack labels, using a controlled testbed environment similar to the one used to generate CICIDS2017.

Online Threshold Adaptation:

The 90th-percentile threshold calibration is computed once at training time and fixed for deployment. In a real network environment, traffic patterns change over time: new applications are deployed, seasonal usage patterns shift, and the mix of protocols evolves. A static threshold that works well at training time may produce increasing numbers of false positives or false negatives as the network baseline drifts. An online threshold adaptation mechanism, where analyst feedback (confirming or dismissing alerts) is used to shift the threshold over time, would substantially improve long-term operational utility. This could be implemented as a simple exponential moving average of the false positive rate, with the threshold adjusted to maintain a target alert rate, or as a more sophisticated Bayesian update mechanism that incorporates analyst feedback as evidence about the current network baseline.

Streaming PCAP Analysis:

The current implementation processes PCAP files as complete batch uploads. Production network monitoring requires real-time or near-real-time analysis of continuous packet streams. A streaming architecture using a ring buffer of recent packets, maintaining running statistics for active flows and emitting flow records when a flow completes or times out, would enable ZeroSight to operate as a continuous network monitor rather than an offline analysis tool. This would require changes to the PCAP parser, the flow aggregation logic, and the dashboard to support streaming updates rather than batch results. The FastAPI server architecture is already compatible with streaming response formats, which would minimize the backend changes required.

Exact SHAP and Feature Interaction Attribution:

The current SHAP attribution approximation, $\text{contribution}_i = |\text{scaled_value}_i| \times \text{RF_importance}_i$, does not capture feature interactions. Two features may individually appear unremarkable but become highly anomalous in combination, a pattern that exact SHAP values (computed via the TreeExplainer algorithm) would capture but the approximation misses. TreeExplainer computes exact Shapley values for tree-based models in polynomial time, making it feasible for per-flow attribution at the throughput of a real network monitor. The primary cost is latency: exact SHAP computation for a 300-tree Isolation Forest and 100-tree Random Forest ensemble would increase per-flow attribution time from under 1 millisecond to perhaps 5-20 milliseconds, which is acceptable for most monitoring use cases. Implementing exact SHAP would also enable cross-feature interaction analysis, identifying pairs of features whose joint deviation is more anomalous than either alone.

Per-Application and Per-Protocol Baselineing:

A single global Isolation Forest trained on all BENIGN traffic treats a CDN bulk download and a DNS lookup as draws from the same distribution. In reality, the statistical properties of these two traffic types are so different that a model trained on their mixture will produce a baseline that is inaccurate for both. A more principled approach would maintain separate IF models for each traffic type, using protocol and application signals from the flow record to route each incoming flow to the appropriate baseline model. This is technically straightforward given the existing four-subtype BENIGN model, but requires a flow classification step (determining whether a flow is web, bulk, interactive, or DNS) before applying the anomaly scorer. This step could itself use a lightweight classifier trained on the same 31 flow features, creating a two-stage architecture: classify traffic type, then score against the corresponding baseline.

Conclusions

ZeroSight demonstrates that a hybrid dual-model architecture combining Isolation Forest and Random Forest, trained on a carefully constructed CICIDS2017-compatible synthetic dataset, can achieve competitive intrusion detection performance with meaningful explainability at inference time. The Isolation Forest achieves 89.9% F1-score in anomaly detection without requiring any labeled attack data during training, providing genuine coverage of novel attacks. The Random Forest provides 94.4% macro-average recall across 18 traffic classes, with per-class SHAP attribution that identifies specific flow features driving each detection. The browser-native PCAP parser and FastAPI server enable deployment without privacy-compromising server-side file upload, making the system suitable for air-gapped and privacy-sensitive environments. Live validation on real browser traffic confirms that DNS and low-rate HTTPS flows are correctly handled, while documenting the burst download false positive as a known limitation with clear XAI-grounded explanation.

The broader picture that emerges from our survey is that network intrusion detection is an increasingly mature field in terms of detection accuracy, but remains immature in terms of operational deployment. The gap between benchmark performance and real-world utility is large and well-documented. Closing this gap requires not only better models but better evaluation methodology: macro-averaged metrics across diverse attack classes, testing on real network captures rather than synthetic datasets, and measurement of analyst response time and accuracy with and without XAI attribution. ZeroSight is a step toward an operationally useful, explainable IDS, but it is a step on a long path. The most important next steps, training on real traffic, adapting to baseline drift, and providing exact feature interaction attribution, require not just technical work but collaboration between ML researchers and practicing security analysts to ensure that what is built is what is actually needed in the field.

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